

Information Retrieval and Search Engines: Introduction

Exercises

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Q1 Consider the following document collection:

- D1** I water the flower that is on the table.
- D2** One chair is behind the table.
- D3** A unicorn is sitting on the chair behind the table.
- D4** The flower that is on the table is now full of water. The chair behind the table is now also full of water.
- D5** The flower is next to the chair that is on the table.
- D6** The chair likes the flower.

Compute the TFIDF weights of the words in the reduced vocabulary $\{water, flower, table, chair, unicorn\}$. You can break down this process by first computing the tf and idf values separately in table 1 and then the $tfidf$ values in table 2.

	tf						idf
	D1	D2	D3	D4	D5	D6	
water							
flower							
table							
chair							
unicorn							

Table 1: TF and IDF values

	D1	D2	D3	D4	D5	D6
water						
flower						
table						
chair						
unicorn						

Table 2: TFIDF values

Based on the results, what are the words that are the most informative for document 4? Would these TFIDF weights be useful in an information retrieval system? 🤔

Q1 🔑 Results are in table 3 and table 4.

Most informative words for D4

water > flower > table > chair > unicorn

Q2 🖋️ Apply length normalization to the previous exercise, just doing that for **D4**. Reflect on the result 🤔.

	tf						idf
	D1	D2	D3	D4	D5	D6	
water	1	0	0	2	0	0	$\log(\frac{6}{2}) = 0.48$
flower	1	0	0	1	1	1	$\log(\frac{6}{4}) = 0.18$
table	1	1	1	2	1	0	$\log(\frac{6}{5}) = 0.08$
chair	0	1	1	1	1	1	$\log(\frac{6}{5}) = 0.08$
unicorn	0	0	1	0	0	0	$\log(\frac{6}{1}) = 0.78$

Table 3: TF and IDF values

	D1	D2	D3	D4	D5	D6
water	0.48	0.00	0.00	0.95	0.00	0.00
flower	0.18	0.00	0.00	0.18	0.18	0.18
table	0.08	0.08	0.08	0.16	0.08	0.00
chair	0.00	0.08	0.08	0.08	0.08	0.08
unicorn	0.00	0.00	0.78	0.00	0.00	0.00

Table 4: TFIDF values

Q2 🗝️ The TFIDF values are given in Table 5, resulting in the following ranking:

term	tf_{ln}	idf	$tf_{ln}idf$
water	1	0.48	0.48
flower	0.5	0.18	0.09
table	1	0.08	0.08
chair	0.5	0.08	0.04
unicorn	0	0.78	0.00

Table 5: TFIDF values with length normalization. $\max_{1 \leq k < l} (tf_{kj}) = 2$

water > flower > table > chair > unicorn

Q3 🖋️ Apply augmented normalized term frequency normalization to the previous exercise, again just doing that for D4. Reflect on the result 🤔. (If you are curious: try varying alpha and observe how that affects the result).

Q3 🗝️ The TFIDF values are given in Table 6, resulting in the following ranking:

term	tf_{aln}	idf	$tf_{aln}idf$
water	1	0.48	0.48
flower	0.75	0.18	0.13
table	1	0.08	0.08
chair	0.75	0.08	0.06
unicorn	0.5	0.78	0.39

Table 6: TFIDF values with augmented normalized term frequency

water > unicorn > flower > table > chair

Q4 🖍 A system has ranked document and the following gives the ranked list of documents where R means relevant and N means not relevant:

$$RRRNRRNRNRRNNNNRNNNNNR$$

Plot the precision-recall graph for this ranking. Where is the breakeven point? What does this graph tell us about the system that produced this ranking? 🤔

Q4 The precision and recall values are given in Table 7.

Example: calculation of recall and precision for the first 4 retrieved documents (RRRN):

1. recall: $3/10$ (3 documents out of 10 relevant documents are retrieved).
2. precision: $3/4$ (3 documents out of 4 retrieved documents are relevant).

Rank	Label	Precision	Recall	Rank	Label	Precision	Recall
1	R	1.00	0.10	13	R	0.61	0.80
2	R	1.00	0.20	14	N	0.57	0.80
3	R	1.00	0.30	15	N	0.53	0.80
4	N	0.75	0.30	16	N	0.50	0.80
5	R	0.80	0.40	17	N	0.47	0.80
6	N	0.67	0.40	18	R	0.50	0.90
7	R	0.71	0.50	19	N	0.47	0.90
8	R	0.75	0.60	20	N	0.45	0.90
9	N	0.67	0.60	21	N	0.43	0.90
10	N	0.60	0.60	22	N	0.41	0.90
11	R	0.63	0.70	23	N	0.39	0.90
12	N	0.58	0.70	24	R	0.41	1.00

Table 7: All precision and recall values.

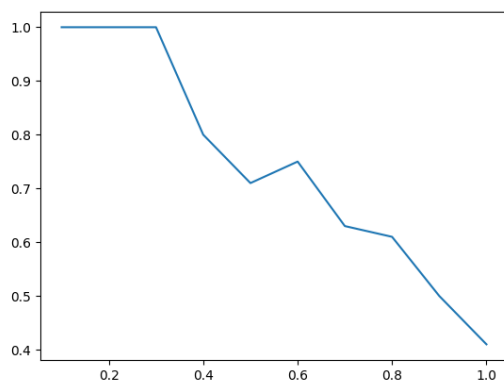


Figure 1: Precision-Recall graph

Recall	Precision	
0.1	1/1	1
0.2	2/2	1
0.3	3/3	1
0.4	4/5	0.8
0.5	5/7	0.71
0.6	6/8	0.75
0.7	7/11	0.63
0.8	8/13	0.61
0.9	9/18	0.5
1	10/24	0.41

Table 8: Precision values for unique recall values.

Instead of plotting all recall and precision pairs given in Table 7, it is common practice to represent them at distinct recall levels. By focusing on unique recall values, we can observe the behavior of the information retrieval system without redundancy. The number of unique recall values corresponds to the number of relevant documents (10 in this case).

The precision-recall graph is given in Figure 1 and the values are presented in Table 8.

Example: calculation of precision when the recall is 0.7:

1. A recall of 0.7 is obtained when 7 out of the top k retrieved documents (the k most relevant documents according to the system we evaluate) are relevant. Knowing that we have 10 relevant documents, the recall is $7/10$.
2. The precision obtained for a recall of 0.7 is $7/k$, where k is the position of the 7th most relevant document (11 in this case). So, precision is $7/11$ (0.63). By selecting k as the position of the 7th most relevant document, we plot the best precision when recall is 0.7.

As we can see from the table and figure, the breakeven point is between 0.6 and 0.7.